



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2006
YEAR 11 MATHEMATICS
HALF YEARLY EXAM

Mathematics (2 unit)

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.

Total Marks – 65

- Attempt questions 1-6

Examiner: *P.Bigelow*

Question 1 (10 Marks)

Marks

- (a) If $f(x) = 1 + x + x^2$, find $f(2)$. [1]
- (b) Expand and simplify $(x + 4)(x - 6)$ [2]
- (c) Express $\frac{1}{\sqrt{2}}$ with a rational denominator [1]
- (d) Factorise [3]
- (i) $a^2 + 3a + 2$
- (ii) $49 - y^2$
- (iii) $x^3 + y^3$
- (e) Solve $7y - 4 = 2y + 11$ [1]
- (f) Evaluate $\frac{7.93}{8.22 - 3.47}$ correct to 3 significant figures. [1]
- (g) If $\sqrt{45} + \sqrt{80} = a\sqrt{5}$, find a . [1]

Question 2 (10 Marks)

(a) Write down the domain of the function $f(x) = \sqrt{1-x}$. [1]

(b) Sketch the line $2x - y - 4 = 0$. [1]

(c) Sketch the parabola $y = (x+1)(x-3)$. [1]

(d) $y = \sqrt{4-x^2}$ is the equation of a semi-circle. [3]

(i) Sketch the semi-circle

(ii) State the domain and range of the function.

(e) Classify the functions as ODD, EVEN or NEITHER. [3]

(i) $f(x) = x$

(ii) $f(x) = x^2 - 1$

(iii) $f(x) = \frac{1}{1+x^2}$

(f) Solve $|x+1| = 2$ [1]

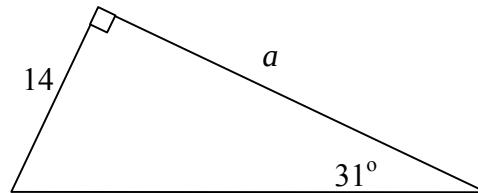
Question 3 (11 marks)

- (a) Solve the following pair of equations simultaneously [2]
$$x - 2y = 4$$
$$2x + y = 3$$

- (b) Find the equation of the line parallel to $x + y - 1 = 0$ and passing through $(4, -6)$. [2]

- (c) Find the angle sum of a regular 10 sided polygon. [1]

- (d) Find the value of a (correct to 1 decimal place). [2]



- (e) Find the value of θ (correct to the nearest minute) where $\sin \theta = \frac{4}{7}$. [2]

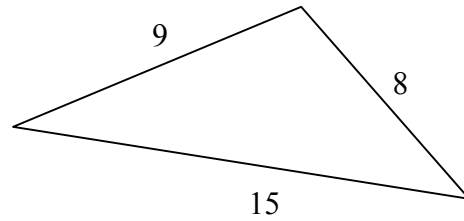
- (f) Find the exact value of [2]

(i) $\sin 60^\circ$

(ii) $\tan 300^\circ$

Question 4 (10 Marks)

- (a) Find the largest angle in the triangle (correct to the nearest minute). [2]



- (b) Sketch $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$ [2]

- (c) Show that $(-4, -5)$, $(2, 7)$ and $(5, 13)$ are collinear. [2]

- (d) Find the shortest distance between the line $4x - 3y + 15 = 0$ and the origin. [2]

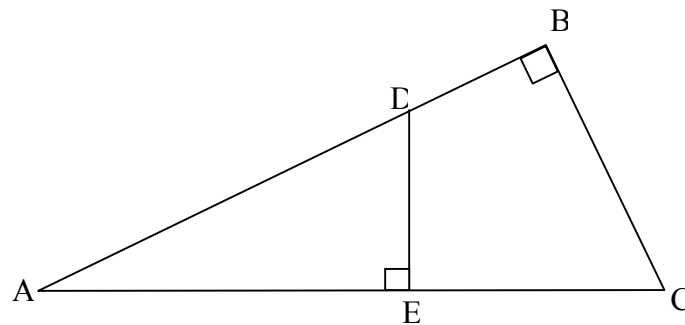
- (e) Find, without a calculator, $0.\dot{1}\dot{5}$ as a fraction in simplest form. [2]

Question 5 (12 Marks)

(a) Write $\sqrt[3]{x^4}$ in index form. [1]

(b) Write down the coordinates of the focus and equation of the directrix for the parabola $x^2 = 20y$. [2]

(c) Show that $\triangle ABC$ is similar to $\triangle AED$. [2]



(d) Given that α and β are roots of the equation $x^2 - 3x + 1 = 0$, find [4]

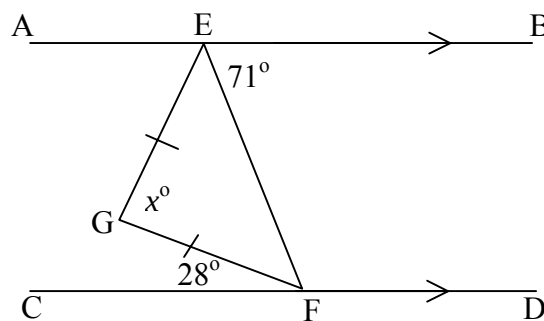
(i) $\alpha + \beta$

(ii) $\alpha\beta$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$

(iv) $\alpha^2 + \beta^2$

(e) AB is parallel to CD. GE=GF. Find the value of x . [3]



Question 6 (12 Marks)

- (a) Find the centre and radius of the circle [2]
$$x^2 - 2x + y^2 + 4y - 4 = 0$$
- (b) Find the equation of the line, passing through the intersection of the lines $2x - y = 1 = 0$ and $3x + y - 6 = 0$ and containing the point $(-2, 3)$. [3]
- (c) A parabola $y = ax^2 + bx + c$ passes through $A(-1, 4)$, $B(0, 7)$ and $C(1, 8)$. Determine the values of a , b and c . [2]
- (d) Given the quadratic expression [2]
$$x^2 + (k - 3)x + k.$$

For what values of k , is the expression positive for all values of x ?
- (e) In $\triangle ABC$, $\hat{A} = 38^\circ 21'$, $b = 11.6\text{m}$ and $a = 7.9\text{m}$. Find the size of the angle B (correct to the nearest degree). [3]

END OF PAPER

2006 Mathematics Continuers: **Solutions Part A**

Question 1 (10 Marks)

- (a) Evaluate $8^{2.1}$ to 4 significant figures,

2

Solution: By calculator, $78.79324245 \approx 78.79$ to 4 sig. fig.

- (b) Write $0.\dot{1}\dot{3}$ as a simplified fraction.

1

Solution: Let $x = 0.\dot{1}\dot{3}$,
 $100x = 13.\dot{1}\dot{3}$,
 $99x = 13$,
 $\therefore x = \frac{13}{99}$.

- (c) Simplify

- (i) $3x - (4 - x)$

1

Solution: $3x - 4 + x = 4x - 4$

- (ii) $\frac{x+1}{3} + \frac{2x}{5}$

1

Solution: $\frac{5x+5+6x}{15} = \frac{11x+5}{15}$

- (d) Convert 270° to radians in exact form.

1

Solution: $270^\circ \times \frac{\pi}{180^\circ} = \frac{3\pi}{2}$

- (e) Factorise

- (i) $x^2 - 9$

1

Solution: $(x+3)(x-3)$

- (ii) $64 + x^3$

1

Solution: $4^3 + x^3 = (4+x)(16-4x+x^2)$

(f) Given that

$$f(x) = \begin{cases} 6 - x^2 & \text{if } x \geq 0, \\ |x| & \text{if } x < 0, \end{cases}$$

evaluate

(i) $f(-2)$

1

Solution: $|-2| = 2$

(ii) $f(0)$

1

Solution: $6 - 0^2 = 6$

Question 2 (11 Marks)

(a) Solve $|2x + 6| = 10$.

2

Solution: $2x + 6 = 10$, or $-2x - 6 = 10$,
 $2x = 4$, $-2x = 16$,
 $\therefore x = 2$. $x = -8$.

(b) Simplify $\frac{1}{\sec^2 \theta} + \frac{1}{\operatorname{cosec}^2 \theta}$

2

Solution: $\cos^2 \theta + \sin^2 \theta = 1$

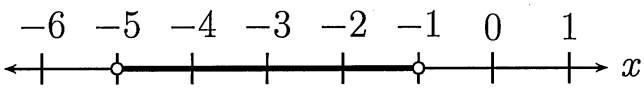
(c) (i) Solve the inequation $|x + 3| < 2$.

2

Solution: $-2 < x + 3 < 2$,
 $-5 < x < -1$.

(ii) Hence graph the solution on a number line.

1

Solution: 

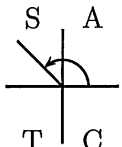
(d) Find the equation of the circle with centre $(-4, 6)$ and radius $\sqrt{5}$.

2

Solution: $(x + 4)^2 + (y - 6)^2 = 5$

- (e) Find the exact value of $\tan \frac{3\pi}{4}$.

2

Solution:  $-\tan \frac{\pi}{4} = -1$

Question 3 (9 Marks)

- (a) (i) Write down the expansion for $\sin(A + B)$.

1

Solution: $\sin A \cos B + \cos A \sin B$

- (ii) Hence find the exact value of $\sin 75^\circ$.

2

Solution: $\sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$
 $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2},$
 $= \frac{1 + \sqrt{3}}{2\sqrt{2}} \text{ or } \frac{\sqrt{2} + \sqrt{6}}{4}.$

- (b) If α and β are the roots of $2x^2 + 3x + 4 = 0$, find the value of

- (i) $\alpha\beta$

1

Solution: $\frac{4}{2} = 2$

- (ii) $\frac{1}{\alpha} + \frac{1}{\beta}$

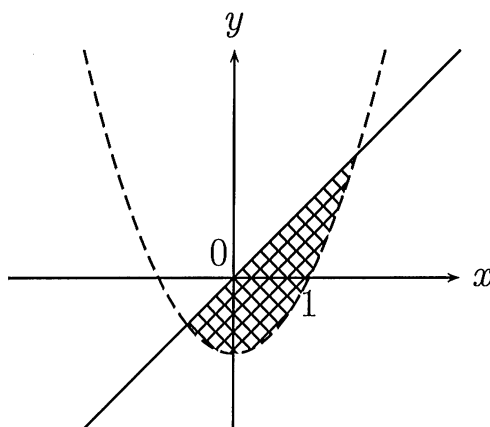
2

Solution: $\frac{\alpha + \beta}{\alpha\beta} = \frac{-3}{2} \times \frac{1}{2},$
 $= -\frac{3}{4}.$

- (c) Sketch the intersection of the regions $y > x^2 - 1$ and $y \leq x$.

3

Solution:



Question 4 (11 Marks)

- (a) State the domain and range of $g(x) = \sqrt{x+4}$.

2

Solution: Domain: $x \geq -4$,
Range: $g(x) \geq 0$.

- (b) Show that $f(x) = x^3 + 3x$ is an odd function.

2

Solution: $f(-x) = (-x)^3 + 3(-x)$,
 $= -x^3 - 3x$,
 $= -(x^3 + 3x)$,
 $= -f(x)$.
 $\therefore f(x)$ is odd.

- (c) Find correct to the nearest minute the angle between $y = 6x - 7$ and $y = x + 3$.

3

Solution: $m_1 = 6$, $m_2 = 1$.
 $\tan \alpha = \left| \frac{6 - 1}{1 + 6 \times 1} \right|$,
 $= \frac{5}{7}$.
 $\therefore \alpha = 35^\circ 32'$.

(d) Find the values of A and B if $2(x - 1)^2 \equiv A(x^2 + 1) + Bx$.

2

Solution: Let $x = 0$, $2 = A$.
Let $x = 1$, $0 = 4 + B$,
 $B = -4$.

(e) For $A(5, 1)$ and $B(-3, 7)$ find the coördinates of the point that divides the interval AB internally in the ratio $3 : 1$.

2

Solution: $\left(\frac{3 \times (-3) + 1 \times 5}{3 + 1}, \frac{3 \times 7 + 1 \times 1}{4} \right) = (-1, 5\frac{1}{2})$

711 CONTINUOUS

SECTION B

5 (a) $x^2 - 6x + y^2 + 10y + 9 = 0$

$$(x-3)^2 + (y+5)^2 = -9 + 9 + 25 \\ = 25.$$

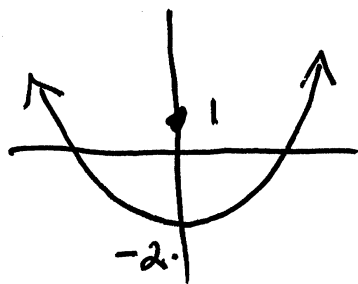
(i) How circle doesn't pass the vertical line test for a function. ✓

OR. There are points (ie two) where the x -value is the same.

OR. It is not a 1-1 correspondence & mapping

(ii) Circle (3, -5) radius 5. ✓✓

(b)



NB $a = 3$.

$$\therefore (x-0)^2 = 4 \times 3 (y+2)$$

$$\boxed{x^2 = 12(y+2)} \quad \checkmark \checkmark$$

(c) $\frac{x-3}{x} < 0$

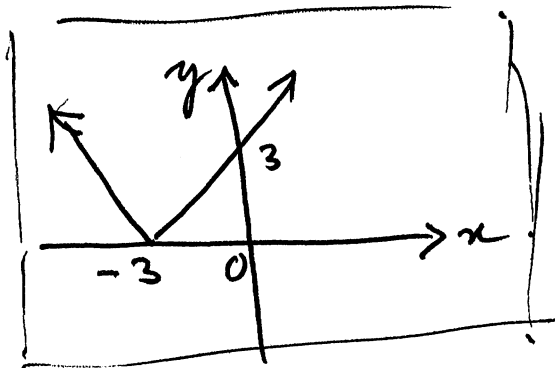
$x(x-3) < 0$ (ie multiply both sides by x^2)

$$\therefore \boxed{0 < x < 3} \quad \checkmark \checkmark$$

(d) $x^4 - 10x^2 + 9 = 0$ let $u = x^2$ $u = 9, 1$ ✓✓✓
 $u^2 - 10u + 9 = 0$ ie $x^2 = 9, 1$
 $(u-9)(u-1) = 0$ $\boxed{x = \pm 3, \pm 1}$

6

(a)



$$(b) \quad y = -a(x-0)(x-4)$$

$$\therefore y = -ax(x-4)$$

now $(2, +4)$ lies on it.

$$\therefore +4 = -2a \cdot -2.$$

$$\therefore \underline{a = 1}$$

$$\therefore \boxed{y = -x(x-4)}$$

$$(c) \quad \text{For real roots } \Delta \geq 0$$

$$\therefore p^2 - 64 \geq 0$$

$$(p-8)(p+8) \geq 0$$

$$\therefore \boxed{p \geq 8, p \leq -8.}$$

$$(d) \quad \tan \theta = \sqrt{3}, \quad 0 < \theta < \frac{\pi}{2}$$

$$\therefore \theta = \frac{\pi}{3}$$

$$\therefore \boxed{\cos 2\theta = \cos \frac{2\pi}{3} = -\frac{1}{2}.}$$

Q7

(a) $4x - 3y - 8 = 0$ and point $(-2, 3)$

$$d = \left| \frac{4x - 2 - 3y - 8}{\sqrt{4^2 + (-3)^2}} \right|$$

$$= \left| \frac{-8 - 9 - 8}{5} \right|$$

$$= \left| -\frac{25}{5} \right|$$

$$= 5$$



(b) Find the intersection of $2x - y = 3$ — (1)
 $3x - y = -2$ — (2)

$$(2) - (1)$$

$$x = -5$$

Sub in (1)

$$-10 - y = 3$$

$$y = -13$$

$$\therefore (-5, -13)$$

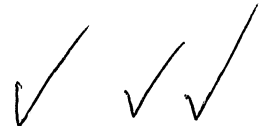
$\therefore$ Equation of line

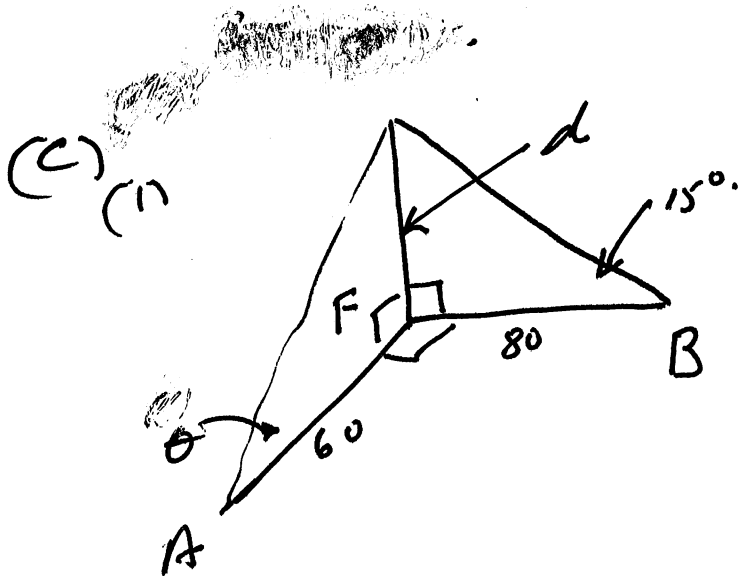
having through $(3, -1)$ and $(-5, -13)$

$$\frac{y + 1}{x - 3} = \frac{-13 + 1}{-5 - 3} = \frac{-12}{-8}$$
$$= \frac{3}{2}$$

$$\therefore 2y + 2 = 3x - 9$$

$$\underline{3x - 2y - 11 = 0}$$





(11) $\frac{d}{80} = \tan 15^\circ$
 $\therefore d = 80 \tan 15^\circ$

now $\tan \theta = \frac{d}{60}$
 $= \frac{80 \tan 15^\circ}{60}$
 $= 0.3573$

$\therefore \theta = 19^\circ 40' \text{ (OR } 20^\circ \text{)}$

(d) $2x + y - z = -3$ — (1)

$-x + 3y - 2z = 1$ — (2)

$x - y + 5z = 12$ — (3)

(1) + (3)

$3x + 4z = 9$ — (4)

(3) $\times 3$

$3x - 3y + 15z = 36$ (3a)

(3a) + (2)

$2x + 13z = 37$ (5)

(4) $\times 2$

$6x + 8z = 18$ — (4a)

(5) $\times -3$

$-6x - 39z = -111$ (5a)

(4a) + (5a)

$-31z = -93$

$z = 3$

Sub in (4)

$3x + 12 = 9$

$3x = -3$

$x = -1$

Sub in (1)

$-2 + y - 3 = -3$

$y = 2$

$\therefore (-1, 2, 3)$

Q8

(a) $x^2 + Lx + M = 0$ has roots $\alpha, 2\alpha$.

Now $\alpha + 2\alpha = -L$ & $\alpha \times 2\alpha = M$

(i)

$\therefore 3\alpha = -L$ — (1)

$\alpha = -\frac{L}{3}$

$2\alpha^2 = M$ — (2)

From (1) & (2)

$2 \times \left(-\frac{L}{3}\right)^2 = M$

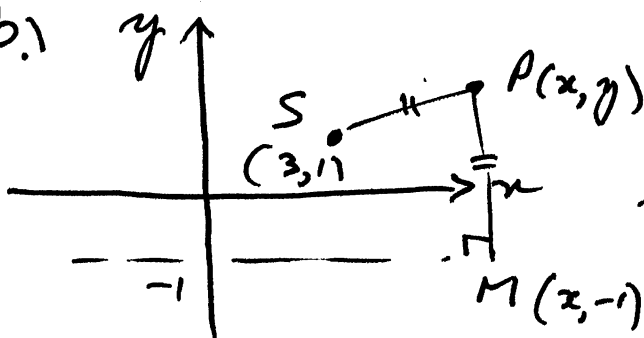
$\frac{2L^2}{9} = M$

✓✓✓

$\boxed{2\alpha^2 = 9M}$

(ii) If L is rational then $-\frac{L}{3} = \alpha$ is rational ✓
 & hence $2\alpha = -\frac{2L}{3}$ is also rational ✓

(b)



Now $SP = PM$

i.e. $\sqrt{(x-3)^2 + (y-1)^2} = y+1$

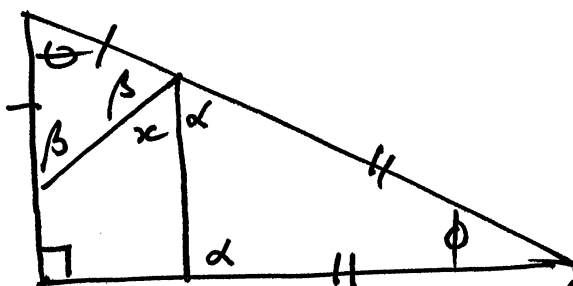
$(x-3)^2 + (y-1)^2 = (y+1)^2$

$x^2 - 6x + 9 + y^2 - 2y + 1 = y^2 + 2y + 1$

$\boxed{(x-3)^2 = 4y}$

✓✓✓

(c)



Now

$\alpha + \beta + \theta = 180^\circ$ — (1)

$\theta + \phi = 90^\circ$

i.e. $(180 - 2\beta) + (180 - 2\alpha) = 90^\circ$ ✓✓

i.e. $2\alpha + 2\beta = 270^\circ$

$\alpha + \beta = 135^\circ$ — (2)

From (1) & (2) $\boxed{\alpha = 45^\circ}$